

2. Consider the following schema for OrderDatabase:

SALESMAN (Salesman_id, Name, City, Commission)

CUSTOMER (Customer_id, Cust_Name, City, Grade, Salesman_id)

ORDERS (Ord_No, Purchase_Amt, Ord_Date, #Customer_id, Salesman_id)

Write SQL queries to

- i. Count the customers with grades above Bangalore's average.
- ii. Find the name and numbers of all salesmen who had more than one customer.
- iii. List all salesmen and indicate those who have and don't have customers in their cities (Use UNION operation.)
- iv. Create a view that finds the salesman who has the customer with the highest order of a day.
- v. Demonstrate the DELETE operation by removing salesman with id 1000. All his orders must also be deleted.

-- CREATE SALESMAN TABLE

```
CREATE TABLE SALESMAN (  
    Salesman_id INT PRIMARY KEY,  
    Name VARCHAR(50),  
    City VARCHAR(50),  
    Commission DECIMAL(5,2)  
);
```

-- CREATE CUSTOMER TABLE

```
CREATE TABLE CUSTOMER (  
    Customer_id INT PRIMARY KEY,  
    Cust_Name VARCHAR(50),  
    City VARCHAR(50),  
    Grade INT,  
    Salesman_id INT,  
    FOREIGN KEY (Salesman_id) REFERENCES SALESMAN(Salesman_id)  
);
```

-- CREATE ORDERS TABLE

```
CREATE TABLE ORDERS (  
    Ord_No INT PRIMARY KEY,  
    Purchase_Amt DECIMAL(10,2),  
    Ord_Date DATE,  
    Customer_id INT,  
    Salesman_id INT,  
    FOREIGN KEY (Customer_id) REFERENCES CUSTOMER(Customer_id),  
    FOREIGN KEY (Salesman_id) REFERENCES SALESMAN(Salesman_id)  
    ON DELETE CASCADE  
);
```

-- INSERT INTO SALESMAN

```
INSERT INTO SALESMAN VALUES
(1000, 'Ramesh', 'Bangalore', 0.15),
(1001, 'Suresh', 'Mysore', 0.12),
(1002, 'Mahesh', 'Bangalore', 0.10),
(1003, 'Ganesh', 'Mangalore', 0.14),
(1004, 'Naresh', 'Hubli', 0.11);
```

-- INSERT INTO CUSTOMER

```
INSERT INTO CUSTOMER VALUES
(2001, 'Arun', 'Bangalore', 300, 1000),
(2002, 'Kiran', 'Mysore', 200, 1001),
(2003, 'Deepa', 'Bangalore', 400, 1000),
(2004, 'Pooja', 'Mangalore', 250, 1003),
(2005, 'Rahul', 'Hubli', 350, 1004),
(2006, 'Sneha', 'Bangalore', 450, 1002);
```

-- INSERT INTO ORDERS

```
INSERT INTO ORDERS VALUES
(5001, 25000.00, '2026-04-08', 2001, 1000),
(5002, 18000.00, '2026-04-08', 2003, 1000),
(5003, 32000.00, '2026-04-09', 2002, 1001),
(5004, 15000.00, '2026-04-09', 2004, 1003),
(5005, 40000.00, '2026-04-10', 2006, 1002),
(5006, 22000.00, '2026-04-10', 2005, 1004);
```

1. Count customers whose grade is greater than Bangalore's average grade

```
SELECT COUNT(*) AS Total_Customers
FROM CUSTOMER
WHERE Grade >
(
    SELECT AVG(Grade)
    FROM CUSTOMER
    WHERE City = 'Bangalore'
);
```

2. Find the name and number of all salesmen who had more than one customer

```
SELECT
    S.Name,
```

```

    S.Salesman_id,
    COUNT(C.Customer_id) AS No_of_Customers
FROM SALESMAN S
JOIN CUSTOMER C
ON S.Salesman_id = C.Salesman_id
GROUP BY S.Name, S.Salesman_id
HAVING COUNT(C.Customer_id) > 1;

```

3. List all salesmen and indicate those who have and don't have customers in their cities (Using UNION)

```

SELECT
    S.Name,
    'Has Customer in Same City' AS Status
FROM SALESMAN S
JOIN CUSTOMER C
ON S.City = C.City

```

UNION

```

SELECT
    S.Name,
    'No Customer in Same City' AS Status
FROM SALESMAN S
WHERE S.City NOT IN
(
    SELECT City
    FROM CUSTOMER
);

```

4. Create a view to display the salesman who handled the customer with the highest order amount on each day

```

CREATE VIEW Highest_Order_Salesman AS
SELECT
    S.Salesman_id,
    S.Name,
    O.Ord_Date,
    O.Purchase_Amt,
    O.Customer_id
FROM SALESMAN S
JOIN ORDERS O
ON S.Salesman_id = O.Salesman_id

```

```

WHERE O.Purchase_Amt =
(
    SELECT MAX(Purchase_Amt)
    FROM ORDERS
    WHERE Ord_Date = O.Ord_Date
);

```

View output:

```

SELECT * FROM Highest_Order_Salesman;

```

5. Delete salesman with ID 1000 and all his orders

```

DELETE FROM SALESMAN
WHERE Salesman_id = 1000;

```

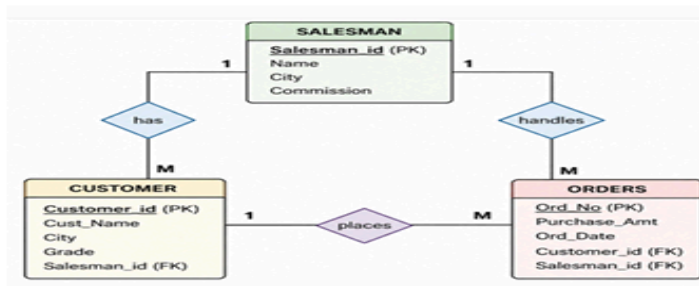


TABLE SCHEMA

Salesman

Column name	Data type	Constraint
Salesman_id	Int	PK
Name	Varchar(50)	Not null
City	Varchar(20)	
Commission	Decimal(10,2)	

Customer

Column name	Data type	Constraint
Customer_id	Int	PK
Cust_name	Varchar(50)	
City	Varchar(50)	
Grade	int	
Salesman_id	Int	FK

Orders

Column name	Data type	Constraint
Order_no	Int	
Purchase-amt	Int	
Oderdate	Date	
Salesman_id	Int	
Customer_id	Int	